Answering Typed Questions with Embedding Distance

Where Cosine Similarity Can Stand in for Label Log-Probabilities, and Where It Cannot

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Abstract

hearim answers Jev-style typed questions (`choice`, `score`, `noul`) by reading the log-probabilities of single-token labels at one decode position. This paper asks whether that call can be replaced by an embedding model: embed the state, embed each option, and turn cosine similarities into a distribution. On one machine we measure three things. First, a published embedding model fine-tuned for typed decisions reproduces its reported accuracy on its own benchmark (0.763), but about twenty points of that come from a trained head, and on seven corpora with unseen schemas the head is worse than the plain cosine of the same encoder every time. Second, plain cosine similarity matches a 27B language model's label log-probabilities on topic classification (AG News, 0.85 against 0.85) and handles a 77-way intent question that the single-token route cannot express, but it collapses on judgments that need reasoning (Korean NLI 0.33 against 0.93; SST-2 0.69 against 0.96). Third, two cheap ways of adapting the embedding route work inside those limits: labeled exemplars used as anchors (Banking77, 0.77 to 0.89 with no training) and fine-tuning the encoder alone (trained schemas gain 10 to 23 points), where training on a single schema family damages the others. The conclusion is that embedding distance is a useful additional route for topic and intent questions and not a replacement for log-probability scoring. hearim now ships it as an opt-in route over five embeddings wire formats.

1 Question

A typed question asks for a distribution over a declared set of options given some state. The companion paper [1] obtains that distribution from a generative model: options are mapped to single-token labels and the answer is the conditional softmax of the label log-probabilities at one position. The method is exact for the model, but each question costs a forward pass of a large model, and the number of options is bounded by the labels that are stable single tokens.

Embedding models suggest a cheaper reading. If the state and each option are mapped into one vector space, the options nearest the state are the likely answers, and distances can be normalized into probabilities. A model built on this idea, fine-tuned for typed decisions [2], reports accuracy above the hosted Jev model on a public benchmark [3]. The practical questions for a gateway that accepts arbitrary schemas are:

1. Does that model's accuracy come from distance, or from what was trained on top of it?
2. How far does distance alone go with a general embedding model, compared with label log-probabilities on the same records?
3. Can the route be adapted cheaply, without the adaptation tying it to one schema?

2 Method

2.1 Scoring by distance

For one question, let x be the instructions followed by the state, and $o_1, \ldots, o_K$ the option texts. With an encoder e producing unit vectors,

$$z_i = \cos\big(e(x),\, e(o_i)\big), \qquad p_i = \frac{\exp(z_i/\tau)}{\sum_j \exp(z_j/\tau)}. \tag{1}$$

The option text is "name: description" for `choice`, the level description for `score`, and the true and false descriptions for `noul`. A structured state is flattened into readable lines (`trace summary · tool errors: 1`) because embedding models read those better than raw JSON. The three primitives are then reduced as on the log-probability route: argmax, probability-weighted level, and $P(\text{true})$.

Two properties follow directly from Eq. (1). There is no label alphabet, so K is unbounded. And p is a temperature-scaled similarity ranking, not a model's belief about the answer: τ decides how sharp it is, and nothing in the encoder was trained to make it calibrated.

2.2 Decomposing the fine-tuned model

The model of [2] scores an option as

$$\text{logit}_i = \underbrace{s \cdot \cos\big(o_i,\ \text{norm}(e(\text{state}) + e(\text{question}))\big)}_{\text{prior}} + \sigma(g) \cdot \big(\text{joint}_i + \text{residual}_i\big),$$

where the second term is a trained head of about 0.5M parameters over projections of the three embeddings. We evaluate the full model, the prior term alone, and Eq. (1) computed with the model's encoder, which is what a client of a standard embeddings API could do.

2.3 Anchors

An anchor is a labeled record: a state whose answer to a question is known. With k exemplar states $s_{i1}, \ldots, s_{ik}$ stored for option i,

$$a_i = \tau_e \log \frac{1}{k} \sum_j \exp\Big(\cos\big(e(s), e(s_{ij})\big)/\tau_e\Big), \qquad z_i \leftarrow z_i + \lambda\,(a_i - \bar{a}), \tag{2}$$

where $\bar{a}$ is the mean over options that have exemplars; options without any are left unchanged. As $\tau_e \to 0$ this is the nearest exemplar, and for large τ_e the class mean. No weights are trained. We use $\tau_e = \tau = 0.05$ and $\lambda = 1$ throughout, without tuning.

2.4 Encoder fine-tuning

The alternative to a head is to train the encoder on the serving objective itself: cross-entropy of Eq. (1) against the gold option, within each question, so the other options act as hard negatives. The result is an ordinary embedding model that any embeddings server can host. We train `embeddinggemma-300m` [4] with AdamW at 2×10^{-5}, batch 16, $\tau = 0.05$.

2.5 Setup

All measurements are from one Apple M2 Ultra.

Corpora. The in-domain benchmark is the test split of `typed-decisions` [3] (400 cases, 2,000 decisions over four workflows). For unseen schemas we draw 150 records each from seven public datasets and phrase them as typed questions: AG News [5] (4-way `choice`), Emotion [6] (6-way), Banking77 [7] (77-way, option names only), SST-2 [8] (`noul`), Yelp stars [5] (5-level `score`), and from KLUE [9] the Korean YNAT topics (7-way) and NLI (3-way). Exemplar pools are drawn from the same sources, disjoint from the evaluation records, at most 16 per option.

Log-probability reference. The same records are scored with the method of [1] on a local Qwen3.8-27B (Q8) served by llama.cpp: options are shown with letter labels and the label log-probabilities at one position are normalized. The calls were made directly against the server with a prompt written for this study, not through hearim's own template; hearim's benchmark command on this particular server fell back to a grammar-constrained path and returned near-chance answers, which is a separate defect.

Embedding models. `embeddinggemma-300m` and `embeddinggemma-2` [4] with their sentence-similarity and retrieval prompts, and `jina-embeddings-v5-text-small` with its text-matching adapter.

3 Results

3.1 The fine-tuned model's accuracy is in its head

Method	Accuracy	choice	noul	score
Uniform (leaderboard)	0.308			
Prior, ignores the input (train mean)	0.478			
embeddinggemma-300m, cosine, classification prompt	0.395	0.218	0.578	0.390
embeddinggemma-300m, cosine, similarity prompt	0.462	0.428	0.580	0.398
Fine-tuned encoder, prior term only	0.551	0.582	0.660	0.448
Fine-tuned encoder, cosine (Eq. 1)	0.573	0.562	0.677	0.504
Fine-tuned model, encoder + head	**0.763**	0.712	0.848	0.739
TypeSafe Jev 1.13.0 (leaderboard, not re-run)	0.727			

Table 1: `typed-decisions` test, 2,000 decisions. The fine-tuned model was trained on this benchmark's train split.

The reported figure (76.4%) reproduces. With the head removed, the same encoder reaches 0.55 to 0.57, and a general embedding model does not beat the input-blind prior. The questions in this benchmark ("does this agent trace need human review?") combine several fields of the state; surface semantic similarity does not answer them.

3.2 Unseen schemas

The trained head does not transfer. On all seven outside corpora the full fine-tuned model is below the cosine of its own encoder, and on SST-2 it is below chance. The head learned the four schemas of its benchmark.

	LLM	Fine-tuned model		General embedding, cosine		
Corpus	logprob	full	cosine	eg-300m	eg-2	jina-v5-small
typed-decisions (training domain)	0.630	**0.748**	0.550	0.458	0.315	0.370
AG News (4)	0.847	0.507	**0.853**	0.647	0.693	0.800
Emotion (6)	0.460	0.373	0.580	0.533	0.567	**0.593**
Banking77 (77)	n/a	0.433	0.740	0.773	**0.780**	0.507
SST-2 (noul)	**0.960**	0.340	0.693	0.633	0.607	0.567
Yelp (score, 5)	**0.627**	0.280	0.493	0.553	0.487	0.360
YNAT, Korean (7)	**0.707**	0.220	0.527	0.440	0.547	0.693
NLI, Korean (3)	**0.933**	0.320	0.327	0.293	0.387	0.280

Table 2: Accuracy, 150 records per corpus (400 decisions for typed-decisions). The 77 options of Banking77 exceed the 26 single-letter labels, so the log-probability reference could not be run.

Distance is good at "what is this about". On AG News the fine-tuned encoder's cosine equals the 27B model. On Banking77, general embedding models reach 0.77 to 0.78 zero-shot over 77 options, a question the single-token route cannot pose at all.

Distance is poor at relation, polarity and synthesis. Every embedding model is at chance on NLI (0.28 to 0.39) where the language model scores 0.933, and SST-2 separates 0.57–0.69 from 0.96. A premise and its contradiction, or a positive and a negative review of the same film, are near each other in embedding space; that is what these models are trained for.

The route is sensitive to model and prompt. The same Jina model with its classification adapter instead of text-matching scores 0.880 on Emotion and 0.167 on YNAT (against 0.593 and 0.693). No single default serves every question.

3.3 Anchors

	Fine-tuned encoder				embeddinggemma-300m			
Corpus	k=0	1	4	16	k=0	1	4	16
typed-decisions	0.472	0.510	0.600	0.578	0.458	0.430	0.470	0.475
AG News	0.840	0.713	0.820	0.833	0.647	0.647	0.780	0.793
Emotion	0.527	0.567	0.533	0.553	0.533	0.500	0.560	0.540
Banking77	0.780	0.907	0.927	0.927	0.773	0.820	0.880	0.887
SST-2	0.667	0.780	0.747	0.720	0.633	0.767	0.780	0.787
Yelp	0.567	0.587	0.573	0.553	0.553	0.500	0.547	0.520
YNAT, Korean	0.407	0.500	0.567	0.567	0.440	0.447	0.613	0.593
NLI, Korean	0.327	0.313	0.327	0.260	0.293	0.307	0.373	0.340

Table 3: Accuracy with k exemplars per option. k=0 is zero-shot. The fine-tuned encoder is rendered here without the model's own text prefixes, so its k=0 column differs slightly from Table 2.

Anchors help where "similar text, same answer" holds. Banking77 rises from 0.78 to 0.91 with one exemplar per intent and to 0.93 with four, and the mean negative log-likelihood falls from 1.97

to 0.45. SST-2 and YNAT gain 5 to 15 points. Emotion and Yelp do not move beyond sampling error, and NLI stays at chance: anchors shift boundaries the encoder already draws and add none. Most of the gain arrives by four exemplars per option.

3.4 Fine-tuning the encoder

Corpus	Base	Specialist	Mixed
typed-decisions	0.463	**0.693**†	0.568†
AG News	0.647	0.647	**0.820**†
Emotion	0.533	0.387	**0.633**†
Banking77	0.773	0.467	**0.853**†
SST-2	0.633	**0.733**	0.707
Yelp	**0.553**	0.407	0.453
YNAT, Korean	0.440	0.540	**0.693**
NLI, Korean	**0.327**	**0.327**	0.287

Table 4: `embeddinggemma-300m` before and after encoder-only fine-tuning, one seed. *Specialist*: 3,000 typed-decisions records, 3 epochs. *Mixed*: 2,392 records from typed-decisions, AG News, Emotion and Banking77, 2 epochs. † marks corpora whose schema was in the training data (records are disjoint from evaluation).

Encoder-only training recovers most of what the head gave in-domain (0.693 against 0.748 on the same 400 decisions) while remaining a plain embedding model. It is not automatically general. The specialist lost 31 points on Banking77 and about 15 on Emotion and Yelp. Mixing four task families removed that collapse, raised every trained schema, and carried over to two of the four corpora it never saw (SST-2 and the Korean topic task, the latter by 25 points from English topic data), while Yelp fell by 10. NLI did not move. A model trained on one schema family should be treated as a specialist and routed by its own name. The model served through hearim reproduced the training script's accuracy exactly (Banking77 0.853, AG News 0.820, YNAT 0.693), confirming that training and serving render the same text.

3.5 Latency

One request with cached option embeddings took 34 to 46 ms (p50) through the embedding route. The log-probability reference took 2.4 to 6.9 s (p50) under an eight-way parallel load on a server shared with other work, so only the order of magnitude is meaningful: about two.

4 What this means for a gateway

Do not replace. Substituting embedding distance for label log-probabilities would cost 30 to 60 points on entailment, sentiment and compound-state judgments. The fine-tuned head is not a way around this: it buys accuracy on the schemas it saw and loses it everywhere else.

Do add. For questions that ask what a text is about, the embedding route can match a 27B model (AG News, with the better encoders), is two orders of magnitude faster, and is unbounded in option count. hearim exposes it as an opt-in route: a model declared as an embedding model is

scored by Eq. (1) through the provider's embeddings API (OpenAI-style, TEI, Ollama, Cohere or Gemini formats), and a request uses it only by naming that model. Nothing in these measurements would justify choosing the route automatically from the question.

Adapt with examples first. Anchors are per-question, reversible, and need one to four records per option where they help at all. Fine-tuning is worth its cost when a schema family is stable and has a few thousand labeled records; it should then be trained as the encoder alone, on a mixture if the model must serve other questions too.

5 Limitations

Each outside corpus has 150 records, so a 95% interval on accuracy is roughly ±7 points, and sampling used one seed; differences under five points in Tables 3 and 4 carry no weight. The log-probability reference is one model with one prompt. Exemplar pools are disjoint from the evaluation records but come from the same source split. Fine-tuning results are single runs. τ_e and λ were fixed, so the anchor figures are closer to a lower bound. Some public datasets may be in the embedding models' training data, which would make their zero-shot figures optimistic; the Jina classification adapter's 0.880 on Emotion looks like such a case, though we did not verify it. Calibration quality was not measured for the embedding route, and `score` questions were trained with hard labels, ignoring the order of levels. The Cohere and Gemini wire formats in the implementation were tested against servers imitating the documented shapes, not against the services.

6 Conclusion

Cosine similarity between an embedded state and embedded options is a legitimate way to answer a typed question when the question is about topic or intent, and it then brings speed and an unbounded option set that single-token scoring lacks. It is not a way to answer questions that require reading a relation between texts or weighing several facts, and no amount of anchoring or encoder training in these experiments changed that. The useful design is therefore both routes side by side, selected by the caller, with labeled examples as the first adaptation tool and encoder-only fine-tuning as the second.

References

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